

Requirement Understanding Document

User Posts and Albums Viewer

Field	Details
Document Version	1.0
Status	Draft - Pending Requirement Clarifications
Application Type	Client-side / presentation-layer web application
Technology	HTML5, CSS3, JavaScript (ES6+), jQuery 4.0.0
Data Source	JSONPlaceholder REST API
Prepared For	Requirement confirmation
Date	02 October 2026

Purpose of this document

This document captures the current understanding of the assignment before implementation. It distinguishes confirmed requirements from assumptions and open questions so that ambiguous points can be clarified before the solution is finalized.

1. Project Overview

The assignment is understood as a small client-side web application that consumes the JSONPlaceholder REST API. The application will retrieve a list of users, present those users through a web interface, and when a user is selected, retrieve and display that user's posts and albums. The application must also provide loading and error handling feedback.

2. Objective

- Retrieve user records from JSONPlaceholder and present them in a usable interface.
- Allow the user to select a specific user from the displayed list.
- Retrieve the selected user's posts and albums using the user ID.
- Present the retrieved posts and albums without requiring a full page reload for each selection.
- Provide clear loading feedback while requests are in progress and appropriate error feedback when requests fail.

3. Scope

3.1 In Scope

- User retrieval from the JSONPlaceholder API.
- Dynamic display of users in the browser.
- User click/selection interaction.
- Retrieval of posts and albums associated with the selected user.
- Dynamic display of posts and albums.
- Loading-state handling.
- API/error handling.
- Basic responsive and presentable styling; no strict visual design has been specified.

3.2 Out of Scope

- Custom backend development.
- Custom database or persistent storage.
- Authentication and authorization.
- CRUD operations against a custom data store.
- Administrative features.
- Advanced UI/UX design beyond a clean, functional presentation layer.

4. Technology and Architecture Understanding

The assignment is understood to be a client-side/presentation-layer website using HTML and jQuery, with CSS for basic presentation. jQuery will handle interaction, AJAX/API communication, response processing, and DOM updates.

JSONPlaceholder is the external data source; no custom backend or database is currently in scope.

Layer / Component	Responsibility
Presentation layer	HTML/CSS structure and visual presentation of users, posts, albums, loading states, and errors.
Client-side application logic	jQuery event handling, API requests, response processing, validation of HTTP success, and DOM updates.
External API	JSONPlaceholder provides user, post, and album data through REST-style HTTP endpoints.

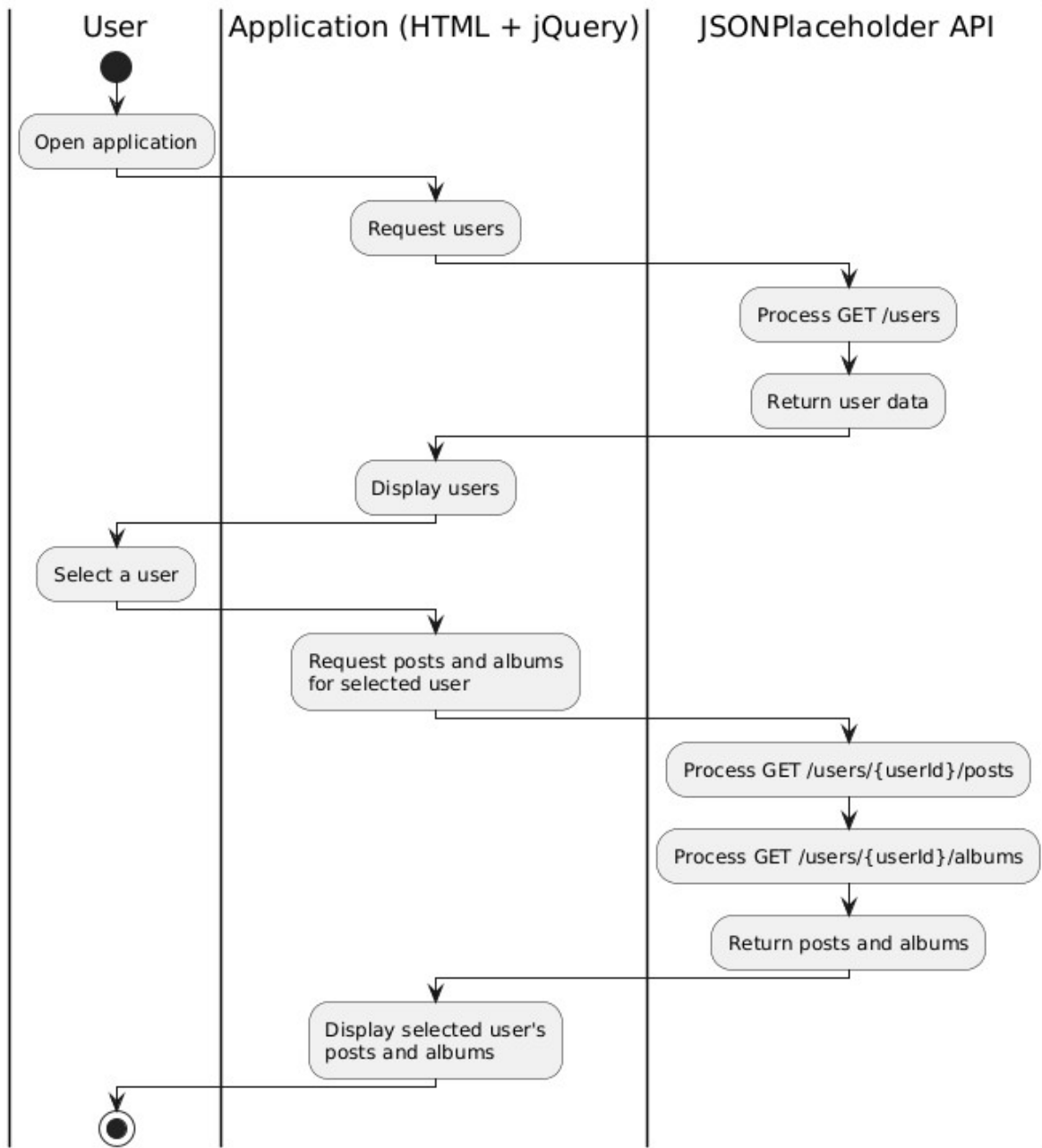
5. Functional Requirements

ID	Requirement	Description	Status
FR-01	Retrieve users	Retrieve users from the JSONPlaceholder /users endpoint.	Confirmed
FR-02	Display users	Display retrieved users; each user must be identifiable and selectable.	Confirmed; displayed fields pending clarification
FR-03	Select user	Allow the user to select/click a displayed user.	Confirmed
FR-04	Retrieve posts	After selection, retrieve posts associated with the selected user ID.	Confirmed
FR-05	Retrieve albums	After selection, retrieve albums associated with the selected user ID.	Confirmed
FR-06	Display related data	Display the selected user's posts and albums.	Confirmed; presentation mode pending clarification
FR-07	Loading state	Show an appropriate loading indication while API data is being retrieved.	Confirmed
FR-08	Error handling	Handle API/request failures with an appropriate user-facing error message.	Confirmed
FR-09	Dynamic update	Selecting another user updates the related data without a manual full-page refresh.	Confirmed by current understanding

6. User Flow

The following requirement flow captures the current understanding of the requested behavior. Detailed technical architecture will be documented later in the HLD/LLD.

User Posts and Albums Viewer - Activity Diagram



7. API Understanding

The following API resources are currently expected to be used:

<https://jsonplaceholder.typicode.com/users> - Retrieve users

<https://jsonplaceholder.typicode.com/users/{id}/posts> - Retrieve posts for the selected user

<https://jsonplaceholder.typicode.com/users/{id}/albums> - Retrieve albums for the selected user

8. Non-Functional Requirements

ID	Requirement	Expected Outcome
NFR-01	Usability	The interface should make user selection and viewing of related data straightforward.
NFR-02	Responsiveness of feedback	The UI should indicate when data is being loaded and avoid leaving the user uncertain about request state.
NFR-03	Error handling	Failures should be handled gracefully with meaningful feedback rather than leaving stale or broken UI.
NFR-04	Maintainability	Client-side code should separate API communication, user interaction, and rendering responsibilities where practical.
NFR-05	Browser compatibility	The application should work in commonly used modern browsers.

9. Assumptions

1. JSONPlaceholder is the sole external data source for this assignment.
2. The application is client-side only and does not require a custom backend or database.
3. A user's ID from the /users response will be used to request that user's posts and albums.
4. The application will dynamically update the current page rather than navigating to a separate page for each user.
5. No strict visual design specification has been provided; a clean and functional layout is acceptable.

10. Open Requirement Clarifications

OC-01 - User fields to display

Please confirm which fields should be shown in the initial user list. Current minimum assumption: the user name. Possible additional fields include email, phone, or other basic profile fields.

OC-02 - Posts and albums presentation

Please confirm whether selecting a user should display Posts and Albums together, or provide separate controls/tabs. Current wording indicates both datasets should be fetched, but the presentation mode is not yet confirmed.

11. Status and Next Action

Current status

Draft prepared based on the current understanding of the assignment. Confirm OC-01 (user fields to display) and OC-02 (whether Posts and Albums appear together or through separate controls). The pending points will be incorporated after requirement clarification, and the confirmed requirements will serve as the basis for subsequent SRS, HLD, and LLD documentation.